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FISHERIES CONFLICT CODEBOOK

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The Fisheries Conflict Database is a collection of reported incidents in which a fisheries resource is contested, disputed, or the source of conflict. The events included in this database will allow us to examine the primary drivers and consequences of fisheries conflict events. The first version of this codebook focuses on countries in Africa.

Using this database, we will address the following research questions:
1. Is fisheries conflict becoming more common or more severe?
2. Is fisheries conflict driven by any of the following variables: illegal fishing, restricted areas, food insecurity, declining fish populations, weak governance, or poor enforcement of fisheries laws?
3. What are the indicators that fisheries conflict will escalate to violent armed conflict?
4. Is fisheries conflict correlated with organized political violence?
5. What role do women play in fisheries conflict (targets of violence, actors of mitigation)?

Fisheries conflict is poorly understood. We use the typology described by Pomeroy et al. (2016)[1] to inform the approach herein. These authors describe a complex 'fish wars cycle' influenced by a suite of linked variables that contribute to a feedback cycle. We isolate several of the key variables described in the fish wars cycle to test the proposed links.

The Fisheries Conflict Database comprises three separate datasets: the **Fisheries Conflict Catalog**, the **Fisheries Dispute Event (FDE) Dataset**, and the **Fisheries Conflict Intensity (FCI) Dataset**.

The Fisheries Conflict Catalog indexes news articles containing information on fisheries conflict. These articles are identified through a systematic search of the Nexis Uni library database by searching named water bodies for disputes or conflict over fisheries resources. Information derived from articles in the Fisheries Conflict Catalog is then populated in the Fisheries Dispute Events and Fisheries Conflict Intensity datasets using the methods outlined in this codebook. The FDE dataset is populated by reported incidents of conflict, dispute, or contestation over a fisheries resource between a minimum of two actors, at a discrete temporal moment, and in a discrete location. The FCI dataset is populated by aggregated FDE entries and other contextual information derived from articles at the scale of waterbody-district-year and aggregated up to the country-year level with assigned levels of intensity.

[1] Pomeroy, Robert; John Parks, Karina Lorenz Mrakovcich & Christopher LaMonica (2016) Drivers and impacts of fisheries scarcity, competition, and conflict on maritime security. *Marine Policy* 67: 94–104.

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Definitions

Terms in bold are further defined in this list.

Fisheries Dispute Event (FDE):
An incident in which a **fisheries resource** is contested, disputed, or the source of conflict between a minimum of two **actors**, at a **discrete temporal moment**, and in a **discrete location**.

Fisheries resource:
A natural aquatic resource (e.g., stocks, populations, species, assemblages) which can be caught by fishing legally. The habitat supporting or containing the resource may also be considered.[2]

Actor:
An individual or group engaging in an event. Groups can be organized or spontaneous. All actors must be people. For example, a hippopotamus attacking fishers for fishing in its territory is not an **FDE** because humans are actors on only one side of the conflict.

Discrete temporal moment:
An event occurs over a defined period of time (days, weeks) versus a systemic condition. For example, a conflict in June 2018 between two fishers over access to fishing gear would meet the definition needed to be counted as an **FDE**, but a description of long-term poverty in a fishing village is not a discrete temporal moment.

Discrete location:
An event occurs in a bounded location (landing site, village, place of business) versus being a diffuse occurrence. For example, conflict at Orion Reef between illegal fishers and marine authorities would meet the definition needed to be counted as an **FDE**, but a general description of dynamite fishing along the coast of Tanzania is not a discrete location.

Fisheries Dispute Aggregate (FDA):
An incident or incidents where a **fisheries resource** is contested, disputed, or the source of conflict between a minimum of two **actors** at a discrete **waterbody-district** level. The primary difference between a **fisheries dispute event** and a fisheries dispute aggregate is the temporal and spatial specificity. An **FDE** requires a discrete beginning and end, while an FDA can contain information over a temporal range that may span multiple years. Likewise, an FDE occurs in an identifiable and local location, while an FDA can occur throughout a larger region or over multiple local locations.

Fisheries Conflict Intensity (FCI):
A qualitatively emergent value regarding level of fisheries conflict at the **waterbody-district-year** unit of observation. This level of analysis creates the opportunity to derive cross-national information regarding the overall levels of fisheries conflict.

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[2] Modified from FAO (1998) Guidelines for the routine collection of capture fishery data. FAO Fisheries Technical Paper 382: 113.

<u>Waterbody-District-Year:</u>

The unit of analysis for **FDAs** is the waterbody-district-year level. This allows for a systematic comparison across cases, and creates clear spatial and temporal boundaries for information regarding fisheries conflict. Many inland water bodies exist entirely in a single administrative level; however, for larger lakes such as Lake Victoria, or rivers such as the Nile, it is necessary to discern at a sub-national level increased spatial specificity of fisheries conflict aggregations. This is extended to coastal regions where the water body remains static but variation exists across administrative levels.

Data Cataloging Procedure

Search Strategy

The Fisheries Conflict Database is constructed from news reporting. We evaluated commonly used and widely distributed databases: Factiva, Keesing's, and Nexis Uni, and we chose the Nexis Uni database for article collection. Nexis Uni contains over 15,000 news sources and has wide temporal and spatial coverage. Furthermore, its predecessor, Lexis/Nexis Academic, historically has been used in academic collection of event data.[3] [4] [5]

We devised a systematically used Boolean search string to cull articles from Nexis Uni. We chose to use water basin, rather than country, as the locating noun in the search string. This increases our chances of finding local-level events, and there is precedent for analyzing at this scale.[6] [7] [8] We use the names of water basins provided by The Source Book for the Inland Fishery Resources of Africa.[9] [10] [11] This is a comprehensive list, by country, of each major water source (rivers, lakes, and reservoirs). We combine one locating noun with a string meant to identify the presence of a fisheries resource. These fishery-based strings are as follows:

> (fish OR fishery OR fisheries OR fisherman OR fishermen OR fisherfolk OR aquaculture OR fishmonger OR mammies OR seine OR cockle)

We do not include words such as 'conflict,' 'dispute,' or 'contested' because this presumes the author has already identified and chosen to write about a conflict or use conflict-specific rhetoric. Additionally, inclusion of these keywords would inflate the returned articles with non-salient articles. The 'fisheries resource' is the least common denominator for an event. Excluding conflict-

[3] Jenkins, J Craig and Doug Bond (2001) Conflict-carrying capacity, political crisis, and reconstruction: A framework for the early warning of political system vulnerability. *Journal of Conflict Resolution* 45(1): 3–31.

[4] Salehyan, Idean; Cullen S Hendrix, Jesse Hamner, Christina Case, Christopher Linebarger, Emily Stull & Jennifer Williams (2012) Social conflict in Africa: A new database. *International Interactions* 38(4): 503–511.

[5] Palmer, Glenn; Vito d'Orazio, Michael Kenwick & Matthew Lane (2015) The MID4 dataset, 2002–2010: Procedures, coding rules and description. *Conflict Management and Peace Science* 32(2): 222–242.

[6] Yoffe, Shim; Aaron T Wolf, & Mark Giordano (2003) Conflict and cooperation over international freshwater resources: Indicators of basins at RISR. *Journal of the American Water Resources Association* 39(5): 1109–1126.

[7] Bernauer, Thomas; Tobias Böhmelt, Halvard Buhaug, Nils P Gleditsch, Theresa Tribaldos, Eivind B Weibust & Gerdis Wischnath (2012) Water-related intrastate conflict and cooperation (WARICC): A new event dataset. *International Interactions* 38(4): 529–545.

[8] Wolf, Aaron T; Shim Yoffe & Mark Giordano (2003) International waters: identifying basins at risk. *Water Policy* 5(1): 29–60.

[9] Vanden Bossche, JP and GM Bernacsek (1990) Source book for the inland fishery resources of Africa: 1. *CIFA Technical Paper* 18(1): 240. Rome: FAO.

[10] Vanden Bossche, JP and GM Bernacsek (1990) Source book for the inland fishery resources of Africa: 2. *CIFA Technical Paper* 18(2): 411. Rome: FAO.

[11] Vanden Bossche, JP and GM Bernacsek (1991) Source book for the inland fishery resources of Africa: 3. *CIFA Technical Paper* 18(3): 219. Rome: FAO.

related terms decreases non-salient articles without missing instances of fisheries conflict. In order to capture fisheries-related conflict occurring in marine or coastal regions that do not have water basin identifiers, we replace in the search term the waterbody identifiers (e.g., 'Lake Victoria,' 'Victoria Lake') with first administrative levels (coastal district names). Given the differing contexts of fisheries conflict around coastlines, we also capture articles that mention coast-specific keywords. To capture all coastal FDEs, we include the following keywords in the search string:

> "exclusive economic zone" OR "territorial waters" OR (coast OR coastline OR coastal OR beach OR island)

This search procedure ensures a broad net is cast to accurately capture fisheries conflicts contained in the Nexis Uni database. Examples of both an inland waterbody and a coastal search string are as follows.

> Inland: (fish OR fishery OR fisheries OR fisherman OR fishermen OR fisherfolk OR aquaculture OR fishmonger OR mammies OR seine OR cockle) AND "lake victoria" OR "victoria lake”

> Coastal: "dar es salaam" OR "mafia marine park" OR "mafia island" OR (muheza OR tanga OR pangani OR bagamoyo OR mkuranga OR rufiji OR kilwa OR zanzibar OR pemba OR lindi OR mtwara OR tanzania OR tanzanian) AND "exclusive economic zone" OR "territorial waters" OR (coast OR coastline OR coastal OR beach OR island) AND (fish OR fishery OR fisheries OR fisherman OR fishermen OR fisherfolk OR aquaculture OR fishmonger OR mammies OR seine OR cockle)

We narrow the search results to 'Africa' in the 'Geography' filter in Nexis Uni, when available. This filter is part of the 'Narrow By' column on the left side of the page. Nexis Uni offers this option when searching for major lakes and coastlines.

The articles returned by the search string are then evaluated for salience for catalog inclusion. News articles are considered salient and included in the Fisheries Conflict Catalog when they contain either specific event information (FDE dataset), or general, aggregate, contextualized information regarding a dispute/conflict between a minimum of two actors over a fisheries resource (FCI dataset).

Fisheries Conflict Catalog

Articles are considered salient for inclusion in the catalog if they meet minimum definition criteria for containing an FDA. A news source must contain information regarding a dispute, conflict, or contested fisheries resource between a minimum of two actors in an identifiable waterbody-district-year. This information does not have to be temporally discrete, meaning it does not have to be a single event occurrence or FDE. By definition, any article that contains an FDE will also include an FDA and therefore will be included.

Catalog Coding Procedure

Salient articles are downloaded and saved as .pdf with bibliography option included (an option in Nexis Uni). The article pdf can be used to extract source-data automatically. Each article is saved as its assigned CatalogID. For example, the article matching CatalogID c1024 is ‘c1024.pdf.’

CatalogID :: [String: c1000+] Unique article index.
Each article should have only one CatalogID—these are index values and therefore do not repeat. CatalogID is assigned sequentially based on the order in which the article was entered into the catalog. Catalog entries begin with c1000 and proceed.

Fisheries Dispute Event Dataset

All articles contained within the Fisheries Conflict Catalog are subsequently coded for both Fisheries Dispute Events (FDEs) and Fisheries Dispute Aggregates (FDAs). If an FDE is coded, at minimum one FDA must also be coded. The beta FDE Google form can be found here.[12]

An FDE is defined as an incident in which a **fisheries resource** is contested, disputed, or the source of conflict between a minimum of two **actors**, at a **discrete temporal moment**, and in a **discrete location** (see Definitions section). After an FDE is identified in a news report, values are derived for each of the variables described below. The coder should assign values based only on information presented in the article and should not use external knowledge or 'read between the lines.' The only exception is that if a previously or subsequently coded article provides new or additional information about the same FDE, an event can be updated to reflect that new information. FDEs can be, and often are, reported by multiple news sources, or by a single news source in multiple articles, which provides updates or evolving information. In such cases, if certain information is contradictory (e.g., the number of arrests during an event), the coder should rely on the most recent report as the source of variable values.

Reminder: some articles may contain information relevant to an FDA but not an FDE, but if an FDE is coded, a corresponding FDA must also be coded. See the Fisheries Conflict Intensity Database section for more information.

Fisheries Dispute Event Coding Procedure

Data Format: Data format is contained in square brackets [ ] following each variable.

- **Binary**: A dummy coding of 0 or 1. Typically, 0 means lack of evidence for or absence of a variable while 1 identifies the presence of the variable (see: **Ongoing**, **WeakGov**, **Marginalization**).
- **Nominal:** A dummy coding corresponding to a category represented by an integer, often used for scales (see: **Intensity**).
- **Ordinal**: An ordered integer (see: **NpartRange**).
- **Category**: Qualitative description of a limited set of options (see: **ActorType**).
- **Count:** An integer that represents the total number of some variable (see: **Fatalities**).
- **String**: Descriptive text (see: **IssueNote**, **LandName**).
- **Date**: Date in DD=day, MM=month, YYYY=year format.

Variables to Code

For each FDE, the coder should record values for the following variables in the FDE Database using the 'Fisheries Dispute Event Form.'

[12] This is an incomplete coding form. It was used on 6 June 2018 to test codebook format and phrasing, and is missing key variables and updates.

Coder :: [Coder Initials: CD]
Indicate who is coding the article by providing the first initial of your first and last name.

CatalogID :: [String: c1000+] Unique article index.
The **CatalogID** value is assigned during cataloging and should simply be copied into the FDE Database. Articles are named by their **CatalogID** (e.g., c1001.pdf), or it can be found in the Fisheries Conflict Catalog in the **CatalogID** column. In the FDE Database, one **EventID** can be derived from multiple **CatalogIDs** (i.e., this cell may have multiple entries).

InfoType :: [Nominal] Relevance of the information provided by the article.

- *1 = Event and Aggregate—Takes coder to Event Section, then Aggregate Section*
- *2 = Aggregate Only—Skips Event Section, takes coder to Aggregate Section*
- *3 = FDE/FDA Relevant Information—Takes Coder to Relevant Information Section*
- *4 = Omit Article—Takes coder to Form Submission Page*

Relevant Information: This section is intended to summarize information in cataloged articles that cannot meet event or aggregate requirements, but contains contextual information (ex: motivations behind conflict drivers). This section is intended for internal use only.

RIWaterBody :: [String] Name of waterbody in cataloged article

RIDate :: [String] Best estimation of the FDE/FDA date based on information provided

RISummary :: [String] Briefly summarize the information that potentially relates to an FDE and/or FDA. This is a crucial variable for future analysis—please prioritize the quality of this variable.

Event Information:

EventID :: [String: e1000+] Unique FDE index

If an event is coded for the first time, assign a new **EventID**. These are unique index values and therefore do not repeat. **EventIDs** are alphanumeric values assigned sequentially based on the order in which the coder first enters each one into the FDE Database. Each unique **EventID** should begin with the letter ‘e’ (denoting Event), and proceed from the first event of e1000.

StartDate :: [Date: MM/DD/YYYY] Date FDE began. If unsure, give best estimate.

CertSDay :: [Binary: 0 or 1] How certain is the coder of the FDE’s start day: unknown/approximated, or stated in the article?

- *0 = Unknown or Approximated*
- *1 = Stated in the article*

CertSMonth :: [Binary: 0 or 1] How certain is the coder of the FDE's start month: unknown/approximated, or stated in the article?

- *0 = Unknown or Approximated*
- *1 = Stated in the article*

CertSYear :: [Binary: 0 or 1] How certain is the coder of the FDE's start year: unknown/approximated, or stated in the article?

- *0 = Unknown or Approximated*
- *1 = Stated in the article*

EndDate :: [Date: MM/DD/YYYY] Date FDE ended. If unsure, give best estimate.
If the FDE is **Ongoing**, **EndDate** should be article publication date.

CertEDay :: [Binary: 0 or 1] How certain is the coder of the FDE's end day: unknown/approximated, or stated in the article?

- *0 = Unknown or Approximated*
- *1 = Stated in the article*

CertEMonth :: [Binary: 0 or 1] How certain is the coder of the FDE's end month: unknown/approximated, or stated in the article?

- *0 = Unknown or Approximated*
- *1 = Stated in the article*

CertEYear :: [Binary: 0 or 1] How certain is the coder of the FDE's end year: unknown/approximated, or stated in the article?

- *0 = Unknown or Approximated*
- *1 = Stated in the article*

Ongoing :: [Binary: 0 or 1] Was the FDE still in progress at the time the article was published?

- *0 = No*
- *1 = Yes*

If the FDE is ongoing, **EndDay**, **EndMonth**, and **EndYear** should be assigned date of article publication and **CertEDay**, **CertEMonth**, and **CertEYear** should be coded as '*unknown or approximated.*' If subsequently coded articles update or clarify the end date, end date variables (and **Ongoing** value, if FDE has ended) should be updated.

CountryName :: [String: Kenya] The name of the country in which the FDE occurred.

- For FDEs occurring in disputed areas (on land or water), code the nation containing the closest named village. If the named village is located in a disputed area, code **CountryName** according to the borders of the UN Geospatial Information Section in the year the event occurred.[13]

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[13] United Nations Geospatial Information Section (http://www.un.org/Depts/Cartographic/english/htmain.htm).

- If a land-linked location is not named, assign **CountryName** to the actor with the closest EEZ (ex: conflict between a Kenyan artisanal vessel and Tanzanian artisanal vessel). If **CountryName** assignment is questionable, explain in the event's **Notes** column.

WBodyName :: [Checklist] Select the name from the list provided. List is based on coding for **CountryName.**

CoastalInfo :: [String] If **WBodyName** is *Coastal* or *EEZ* provide details (ex: 20 nm off coast of Dar es Salaam). If an inland waterbody (lake, river, etc.), leave answer blank.

SideA1Type / **SideA2Type** :: [Category] Type of actor(s) on Side A of conflict.
SideB1Type / **SideB2Type** :: [Category] Type of actor(s) on Side B of conflict.
Each FDE will have two sides—A and B. A/B sides are agnostic regarding 'who started it.' Side A and Side B must be in opposition, but the designation A versus B is meaningless. If a side involves only one actor, leave the corresponding '2' columns blank. For **SideAType** and **SideBType**, choose from the following possible types:

- *Domestic Fishers*
- *Foreign Fishers*
- *Fishing Village*
- *Security Forces —Military*
- *Security Forces—Police*
- *Security Forces—International (ex: multinational patrol groups, NATO)*
- *Security Forces—Resource (ex: marine park officials, rangers)*
- *Security Forces—Local (any local but non-police patrol groups)*
- *Government—Local (anything lower than 1st Administrative level)*
- *Government—Regional (1st Administrative level)*
- *Government—Federal*
- *Government—International*
- *Fishing Collective (ex: Beach Management Units (BMU), fisheries union or cooperative)*
- *Tourism (ex: hotels, tourists, SCUBA companies)*
- *Rebels—Organized (durable rebel groups, militias, terrorist organizations; group should have a name, ex: Al Shabaab)*
- *Rebels—Other (Unnamed 'militants,' 'insurgents,' or 'guerrillas'), e.g., the article refers to the group as rebels or there are political motivations to the group's existence; for example, rebels might steal fishing boats for use in moving members or trafficking arms*
- *Bandits—Criminals or actors motivated by financial gain or non-political motivations such as stealing boat motors for re-sale and profit*
- *Other (not included in any category above)*

SideA1Name / **SideA2Name** :: [String] Name(s) of actor(s) on Side A of conflict. Ex: Ugandan People's Defense Force, President Museveni.
SideB1Name / **SideB2Name** :: [String] Name(s) of actor(s) on Side B of conflict. Ex: local village elders, Boko Haram.

NpartExactA1/A2 :: [Count] Exact count of Side A participants. If number is not provided, select 'unknown.'
NpartExactB1/B2 :: [Count] Exact count of Side B participants. If number is not provided, select 'unknown.'

A/BWoman :: [Binary: 0 or 1] Does the article contain an indicator of at least one female actor on Side A or B? Indicators include feminine pronouns or other vocabulary that indicates the involvement of a person identifying as female: for example, she, her, Miss, Mrs., Ms., women, woman, female(s), lady(ies), girl(s). This variable should not assume women are participants in groups (villagers, residents, shoppers, fishermen). The article must provide evidence of female involvement.

- *0 = No women were involved in Side A/B*
- *1 = There was at least one woman participant in Side A/B*

Example: Boko Haram attacks a fishing village = **SideAType**: Rebels - Organized, **AWoman**: 0; **SideBType**: Fishing Village, **BWoman**: 0
Example: The Tanzanian and Malawian presidents are disputing their border on Lake Malawi = **SideAName**: Tanzanian President Jakaya Kikwete, **AWoman**: 0; **SideBName**: Malawian President Joyce Banda, **BWoman**: 1

NpartRange :: [Ordinal: 1:4 or -99] Coder's best estimate of the total population involved in (not affected by) the FDE on both sides.

- *1 = 1–10 people*
- *2 = 11–100 people*
- *3 = 101–1,000 people*
- *4 = 1,001+ people*

Example: 20 guerrillas attack a 250-person fishing village = 3
Example: The Forces for the Defence of Democracy, a Hutu opposition group in Burundi, accused the Burundi government of using a fishing ban to enrich themselves and starve lakeside fishing communities = -99
Example: A collective of local fishers petitions the regional fisheries minister to change the type of net banned in the area. We assume the collective is small = 2

NpartExact :: [Count] Exact count of population involved in the FDE. If unknown: -99
Example: 20 guerrillas attack a 250-person fishing village = 270
Example: The Forces for the Defence of Democracy, a Hutu opposition group in Burundi, accused the Burundi government of using a fishing ban to starve lakeside fishing communities = -99
Example: A collective of local fishers petitions the regional fisheries minister to change the type of net banned in the area = -99

FishType :: [Category] The type of fish involved in the FDE. We have highlighted two specific types of fishes (sharks and tuna) that have important commercial and ecological value and may be of interest to users.

- *Inland—Lake, River, and Reservoir events, even if specific fish type is not mentioned*
- *Coastal—Coastal fishes excluding tuna or sharks, including non-finfish resources (coder hints include fishing close to shore, small boats, use of dynamite or gillnets or diving gear)*
- *Tuna (not conclusive but coder hints include marine fishing far from shore, longlines or purse seine gear, foreign vessels (not always), mention of Regional Fisheries Management Operations)*
- *Sharks (not conclusive but coder hints include longline or gillnet fishing, finning)*
- *Other (identified fish that do not fit into inland, coastal, tuna, or sharks categories)*

IllegalFishing :: [Nominal: NA or 1:5] Was illegal fishing involved in the FDE? Laws vary depending on location. The article must state that the gear, location, or species targeted were illegal in the country under consideration.

- *NA = No illegal fishing*
- *1 = Gear (fishing with banned gear (ex: trawlers) or nets that have small mesh sizes)*
- *2 = Location (fishing in closed areas (ex: marine parks) or during closed seasons)*
- *3 = Species (catching endangered species (ex: turtles, mammals, whale sharks))*
- *4 = Unlicensed (fishing without a state flag or license, foreign fishing in national waters without permission)*
- *5 = Other (No radar, no electronic tracking device, flagless, banned female fish)*

Conflict Drivers

The following 14 variables are indicators of the drivers of the conflict, including motivations of the actors and enabling factors or conditions. When coding in this section, the coder should ask themselves the question ‘Were the actions of the actor(s) involved in the FDE driven by (<u>insert indicator)</u>?’

Conflict drivers should be assessed from the information provided in the pertinent article—do not make assumptions or apply external or prior knowledge of the situation. An FDE can be coded for multiple drivers.

1. WeakGov :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by weak governance?

- *0 = No*
- *1 = Yes*

<u>Examples</u>:

- Corruption, weak enforcement (security or judicial), weak institutional capacity (inefficient legal framework, conflicting authorities), lack of public participation, inadequate information, organized crime.
- A regional government holds an emergency meeting to deal with dynamite fishing after a deluge of complaints from coastal fishing villages. Complaints include rampant corruption among local government officials and a lack of accountability. The fisheries minister

stated that the goal of the meeting should be to close up the numerous loopholes in their legal framework.
- The borders of a marine park are not patrolled due to lack of funding from the government to the security forces. Violations of the park boundaries result in interactions and conflict between fishers and other users of the marine park, such as tourist SCUBA divers.

Not Included: Corruption or weak enforcement that is not directly linked to the actors' motivations.

2. FishPop :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by an actual or perceived decline in fish population(s)?
- *0 = No*
- *1 = Yes*

Examples:
- Actions taken in response to unsustainable catch rates/quotas or overcapacity of licensed fishing fleets.
- Motivation of actors to save dwindling fish populations or replenish fish stocks.
- Other coded motivations (e.g., **FoodInsecurity**) are related to declining fish populations.

Not Included: Events not directly motivated by concern for the fish stocks.

3. EcoChngOther :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by changes to the natural ecosystem, excluding health of fish populations?
- *0 = No*
- *1 = Yes*

Examples:
- Ecosystem changes—eutrophication, ocean acidification or temperature change, algal blooms, sea level change, drought, loss of biological diversity, land-based pollution, or marine debris.
- Coral reef damage (common in dynamite fishing cases).

Not Included: Changes to fish populations.

4. Poverty :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by conditions of poverty, excluding food insecurity?
- *0 = No*
- *1 = Yes*

Examples:
- Limited livelihood options, lack of public health services, or lack of public education services.
- Fishers are upset about a fishing ban because they have no other way of making a living.

Not Included:
- Food insecurity motivations (threat of famine).
- Poor fishers fighting over gear (unless the article states that their poverty motivated their actions; ex: the actors could not afford gear, so they had to share).

5. FoodInsecurity :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by food insecurity? Food insecurity is a lack of access to a reliable source of sufficient and nutritious food (both fisheries and non-fisheries food).

- *0 = No*
- *1 = Yes*

Example: Fishing ban results in declines in fish availability and nutrition in a remote fishing village and conflict over access to agricultural land or grazing animals results.

Not Included: Actions motivated by dwindling fish stocks. Do not assume fish are the only food source of the motivated actor unless explicitly stated. Such a case would receive a 1 for **FishPop**.

6. Marginalization :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by social, economic, ethnic, tribal, gender, or political marginalization?

- *0 = No*
- *1 = Yes*

Example: Congolese militia targets ethnic Tutsis in a fishing village on Lake Tanganyika, killing over 50.

Not Included: The event does not mention either of the actors being targeted for their ethnic, tribal, gender, or political identity.

7. GroundsLim :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by limitations on access to fishing grounds?

- *0 = No*
- *1 = Yes*

Examples:

- Fishing where the actor shouldn't be fishing, in the opinion of at least one of the actors (ex: border conflict, unlicensed foreign fishers, marine parks, protected areas).
- Non-fisheries users, contested maritime boundaries, closed areas/reserves, limited access areas.
- All **Disputed** areas will most likely be **GroundsLim**, but not all **GroundsLim** will be **Disputed.** If the actors violating access limitations are doing so in stealth, it is not disputed (they acknowledge the limitation). If the access limitation is contested (e.g., in the press, the courts, or legislature), it is disputed.

Not Included: Two groups of fishers from different ethnic groups get into a fight while fishing in a protected marine park. Both groups were illegally fishing in the marine park, but the event was motivated by ethnic tensions, not the fishing grounds.

8. OpsScales :: [Binary: 0 or 1] Was at least one of the actors motivated by competition with actors that operate at a different scale of fishing? Scale of fishing refers to the sector (commercial/industrial versus artisanal/subsistence), which can usually be determined by the size of boat, number of crew, flag of vessel, gear, or location of fishing. Possible terms include: subsistence, small-scale, artisanal, recreational, commercial, large-scale, industrial.

- *0 = No*
- *1 = Yes*

Examples:

- Tanzanian fishers complain because the Tanzanian government issued a trawling license to a commercial Kenyan trawler vessel but turned down artisanal Tanzanian trawling applications.

Not Included:

- Domestic small-scale fishers fighting other domestic small-scale fishers.

9. ForeignFisher :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by the presence of foreign fishers in domestic waters?

- *0 = No*
- *1 = Yes (If 'Yes,' then provide answer for **ForeignFisherTxt** in section following drivers)*

Examples:

- Fishers attack foreign fishers who entered their waters.
- Security forces arrest unlicensed foreign fishers.
- Domestic fishers are angry at their government for approving licenses to foreign fishers.

Not Included:

- Foreign fishers are mentioned, but are not a motivating factor in the FDE actor's actions.

10. Markets :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by the supply or demand from transnational markets? Do not assume the presence of a foreign boat represents high demand in that country. There must be something about the state of the foreign market.

- *0 = No*
- *1 = Yes*

Examples:

- A foreign government bans imports from the FDE's country or body of water.
- Using illegal fishing methods to meet unusually high demand in another country.
- Fishermen crossing borders in search of better prices or to sell fish to a different population.

Not Included:

- The crew of an unlicensed Chinese vessel found in Tanzanian waters were arrested. No mention of the demand for fish in China.
- Events that only deal with the state of the FDE's market.

11. IncrEfficiency :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by increased efficiency of fishing by the other actor(s)?

- *0 = No*
- *1 = Yes*

Examples:

- Destructive fishing practices (illegal) that collect fish rapidly and in high volumes, or highly efficient gear types (legal).
- Technological advances such as fish-finders or sonar, and changes in fishing gear (e.g., smaller net mesh sizes) aimed at increasing catch.

Not Included: A dispute that involves the use of the gear listed above, if the gear is not a motivating factor for either actor's engagement.

12. IncrPressure :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by conditions related to increased fishing pressure?

- *0 = No*
- *1 = Yes*

Examples:

- Increased market demand for seafood in a country, or increased number of fishers at a waterbody.
- Human population growth.
- Influx of refugees causes the fishery to collapse (would also be coded 1 for **FishPop**).

Not Included: Articles that mention an influx of migrants and a fishery dispute, but do not explicitly link the two.

13. Crime :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by a non-fisheries maritime crime? Event does not need to be marine. Inland bodies of water apply.

- *0 = No*
- *1 = Yes*

Examples: Smuggling, trafficking, piracy, abductions or armed robbery against people (including fishers) or cargo. FDE is considered an abduction if actors ask for a ransom.

Not Included: 'Piracy' of fish (this is IUU fishing).

14. StratLoc :: [Binary: 0 or 1] Were the actions of at least one of the actors driven by the strategic importance of the fishery's land location? Not related to access to fish.

- *0 = No*
- *1 = Yes*

Examples: Rebel groups, tourism, competition over shoreline location.

Not Included: Events over fishing grounds.

[End of Conflict Driver Variables]

ForeignFisherTxt :: [String] If answer to **ForeignFisher** is 'yes,' who were the foreign fishers in this event's domestic waters? (Ex: Kenyan trawlers, Tanzanian fishers)

ViolenceScore :: [Ordinal: 1:3] Each FDE is assigned a value for intensity. Select the highest level met. For example, if boats are stolen (mid-level) and people are killed (high level), the event would receive an intensity score of 3.

- *1 = Low-Level Intensity—Conflict is verbal, no physical action is taken*
    - *Bans*
    - *Complaints*
    - *Fines*
- *2 = Mid-Level Intensity—Action is taken, but no physical harm is done*
    - *Property damage*
    - *Stolen property*
    - *Gear confiscation*
    - *Arrests*
    - *Abductions*

- *3 = High-Level Intensity—Event involves physical harm*
    - *Injuries*
    - *Sexual assaults*
    - *Fatalities*

Fatalities :: [Count] How many fatalities were part of the FDE? Record exact number if known.
- -99: unknown (article must mention that fatalities occurred)
- -88: unknown, but probably small (less than 10)
- -77: unknown, but probably high (more than 10)

Injuries :: [Count] How many injuries, excluding sexual assault, were part of the FDE? Record exact number if known.
- -99: unknown (article must mention that injuries occurred)
- -88: unknown, but probably small (less than 10)
- -77: unknown, but probably high (more than 10)

SexualAssault :: [Count] How many sexual assaults, including rape, were part of the FDE? Record exact number if known.
- -99: unknown (article must mention that sexual assault occurred)
- -88: unknown, but probably small (less than 10)
- -77: unknown, but probably high (more than 10)

Arrests :: [Count] How many arrests were part of the FDE? Record exact number if known. Deportations are included in this category.
- -99: unknown (article must mention that arrests occurred)
- -88: unknown, but probably small (less than 10)
- -77: unknown, but probably high (more than 10)

Abduction :: [Count] How many abductions were part of the FDE? Record exact number if known.
- -99: unknown (article must mention that abductions occurred)
- -88: unknown, but probably small (less than 10)
- -77: unknown, but probably high (more than 10)

PropDamage :: [Nominal: 0:3] The level of property damage in the event was:
- *0 = No property damage*
- *1 = Minor property damage or confiscation*
- *2 = Confiscated, damaged or stolen boats, engines, or other larger equipment*
- *3 = Damage to buildings, homes, or entire villages*

FemaleEvent :: [Binary: 0 or 1] Does the female identity play a significant role in the event?
- *0 = No*
- *1 = Yes*

<u>Example</u>: The main actor(s) organized on the basis of female gender identity.
<u>Example</u>: The target(s) are explicitly female.
<u>Example</u>: Motivating issue(s) centers on matters of specific concern to women.

MaleEvent :: [Binary: 0 or 1] Does the male identity play a significant role in the event?

- *0 = No*
- *1 = Yes*

Example: The main actor(s) organized on the basis of male gender identity.
Example: The target(s) are explicitly male.
Example: Motivating issue(s) centers on matters of specific concern to men.

CrossNational :: [Binary: 0 or 1] Did the FDE include more than one country? The dyad of actors must contain a minimum of two nationalities in order for the FDE to be considered cross-national.

- *0 = No*
- *1 = Yes*

LandName :: [String] Name of the closest land-based location. This variable will be used for geocoding our events, so try to use something that can be searched in Google Maps, such as the town or village reported. If necessary, a higher jurisdiction like region or district can be used. Each event should have only one location. If a land-based location is not provided (EEZ or High Seas events), include distance from shore; ex: '100 nm off Tanzanian coast.'

LocAccuracy :: [Nominal: 1:4] The specificity of event location provided by the article.
Location on shore is preferable for understanding conflict around large fisheries. Ambiguous coastal FDEs should be coded at the 'Country' level for accuracy.

- *1—Town*
- *2—District (1st administrative level)*
- *3—Region (General region of FDE is listed, but no town or district; ex: Northeastern Kenya)*
- *4—Country*

Disputed :: [Binary: 0 or 1] Did the event take place in a disputed area?
A disputed area is one in which actors are fishing where their presence is not uniformly recognized as legitimate. That is, some actors say those fishers should not be in the area, while the fishers claim they can.

- *0 = Area was not disputed*
- *1 = Area was disputed*

Example: An island on the border of Uganda and Kenya in Lake Victoria is disputed at the International Court of Justice. Fishers on both sides claim a right to fish from the island.
Not Included: Fishers violate boundaries of closed areas or limited access areas but do so knowing they are violating laws or regulations and may do so in stealth.

IssueNote :: [String] Briefly summarize the event in 1–2 well-constructed sentences. Please include justification for selections made, such as **Ongoing** = Yes or Actors listed. This is a crucial variable for future analysis—please prioritize the quality of this variable.

Notes :: [String] For internal use only. Notes of interest from the coder, such as interesting cases, concerns with dates, ideas for analysis, etc.

Aggregate Catalog

FDEs will contribute to the creation of the Fisheries Conflict Intensity dataset. This requires aggregating FDEs with other contextual information about conflict that does not meet the definition of an FDE. Therefore, generalized data about fisheries conflict occurring at a larger spatial and temporal scale is extracted. The aggregation of events and contextualized information proceeds after the FDE dataset is completed for a given country. While coding FDEs, aggregate waterbody-district-year information is cataloged.

As a reminder, all FDE events generate an FCI dataset entry, but not all FCI entries correspond to an event (i.e., some FCI entries are derived from the generalized contextual information described in the paragraph above). This means that every coded FDE needs to have this section completed. If an article does not contain an FDE but does have 1+ aggregates, we record the **CatalogID** and code the following four variables for each aggregate. The coding form provides the option to enter information for as many aggregates as needed.

AggCountryName :: [Category] The country where there aggregate took place.

AggYearStart :: [Date: YYYY] The year the aggregate began.

AggYearEnd :: [Date: YYYY] The year the aggregate ended.

WBDis :: [Category - Check List] The district of the waterbody where the aggregate occurred. If multiple districts, please select each that applies.

AggSummary :: [String] A brief summary of the aggregate. If your **AggSummary** will be the same as your **IssueNote** (meaning the article does not provide information about a larger aggregate beyond the FDE), leave this response blank.

<u>Updating Event Entries</u>

If multiple articles inform the FDE, the dataset entry can be edited/updated to reflect new information. For example, an article (c1065) describing e1025 reports that fishers were arrested on a reservoir for fishing at night with banned nets. At the time of c1065's publication, the event was ongoing. A more recent article (c1092) mentions e1025, including 'the 6 fishermen were arrested in the early hours of the morning for night fishing on the reservoir, where night fishing had been banned, and for their use of illegal gear.' The new article updates the time of the FDE (and therefore the StartDay) and the number of arrests, and provides new knowledge of illegal activity and describes ground limitations. The article that provides additional information should be added to **CatalogID**.

Geocoded Variables in FDE Analysis

When FDE coding has been completed for a country, all FDEs will be geocoded. Automatic geocoding will be conducted using the Geocoder[14] module. The **LandName** coded in the FDE dataset will be combined with the **CountryName** to form a tuple in the format of ('location, country') which will then be fed to the geocoder. Once the geocoder has sufficiently located the point of interest, longitude and latitude coordinates will be extracted and stored added to the FDE dataset. An automated script will extract location name for each event and assign a latitude and longitude coordinate using Google Earth's API. Geo-coordinates will be derived manually for location names that do not automatically map using the API.

Fisheries Conflict Intensity Dataset

The Fisheries Conflict Intensity (FCI) dataset uses the 'Aggregate Catalog' data collected in the FDE. The FCI documents the intensity of fisheries conflict in a given waterbody-district-year (referred to as a Fisheries Dispute Aggregate or FDA). This intensity level can then be aggregated to the country level to facilitate cross-national analysis. This dataset will result in a qualitatively emergent value regarding the overall level of fisheries conflict at the state-year unit of observation. The purposes of this dataset are: 1) to scale events up to a unit of analysis compatible with explanatory variables of interest (e.g., national fishery stock status, socioeconomic metrics, other forms of armed conflict); 2) to assess 'conflict intensity' across all events in a country-year; and 3) to incorporate information derived from contextual information that is not well captured by documented FDEs. In this sense, the combined FDE-FCI dataset pair is similar to the Uppsala Conflict Data Project's Georeferenced Event Dataset paired with the UCDP Armed Conflict Dataset.

AggIntensity :: [Ordinal: 1:3] Level of fishery dispute conflict intensity, from the perspective of fisheries stakeholders.

- 1 = Low (Inconvenience); ex: complaints about government officials, action motivated by government trying to make money, local bans without a response from fishers.
- 2 = Medium (Serious Problem or Threat); ex: piracy of gear, complaints that lead to government action, forced migration, arrests and gear confiscation without law enforcement violence or harassment.
- 3 = High (Fearful Fishers); ex: piracy including abductions or murder, abductions or arrests over border disputes, arrests and gear confiscation with attendant violence or repression.

[14] Geocoder (http://geocoder.readthedocs.io/).